Lecture 6: Pterosaurs

Pterosaurs were the first vertebrates to evolve powered flight. They were contemporaneous with the non-avian dinosaurs, appearing towards the end of the Triassic Period and going extinct at the end of the Cretaceous. Unlike their dinosaurian cousins (they were not dinosaurs themselves), pterosaurs left no living descendants which, together with their unique adaptations for an aerial existence, has made them difficult to reconstruct. Their bizarre morphology was in fact instrumental in alerting comparative anatomist Georges Cuvier and others at the turn of the nineteenth century that the lifeforms of the ancient past might be wholly different from those we see today.

Pterosaur anatomy

Pterosaur wings were membranous, and were supported by the enormously elongate fourth finger. Pterosaurs also possessed a hollow skeleton, a unique bone called the **pteroid**, a broad sternum with an anterior **cristospine** (origin of the main downstroke muscles), a large **deltopectoral crest** on the humerus (insertion of the downstroke muscles), and unique **pre-pubic** bones. The pteroid articulated at the wrist and supported a membranous forewing. Most of these characters relate to flight. The pre-pubic bones were probably involved with respiratory movements: **pneumatic openings** in the bones indicate that pterosaurs had an air sac system rather like that of birds. Basal pterosaurs (sometimes known as the paraphyletic **'rhamphorhynchoids'**) had a clawless fifth toe and, usually, a long, variously stiffened tail bearing a small, vertically-oriented membrane at the tip. The short-tailed **anurognathids** were notable exceptions, and were also the only basal pterosaur group to survive into the Cretaceous. It is in this group that we have some of the best evidence that pterosaurs were endothermic: filamentous integumentary structures that bear a striking resemblance to simple feathers. The largest basal pterosaur known was the rhamphorhynchid *Dearc*, from the Middle Jurassic of Skye, with a reconstructed wingspan between 2m and 3m.

The derived **pterodactyloid** pterosaurs arose towards the end of the Jurassic Period and were the principal pterosaurs of the Cretaceous. They had an elongate neck and skull, often with reduced or absent teeth, a shortened tail, a reduced/absent fifth toe, a single **nasoantorbital fenestra**, and an elongate metacarpus and pteroid. The Middle Jurassic *Darwinopterus* shows that the changes in the skull and neck likely preceded those in the rest of the postcranium. Early pterodactyloids, like *Pterodactylus* itself, were modest-sized (1m-1.5m adult wingspan), but during the Cretaceous the group showed increased morphological disparity, and generally increased in size. Notable groups were:

- The **ctenochasmatids**, with obvious filter-feeding adaptations.
- The more morphologically conservative **ornithocheirids**, that included the largest toothed pterosaurs, possibly reaching wingspans of 8m.
- The toothless **pteranodontids**, with their famous (and sexually dimorphic) backward-projecting head crest, that reached similar sizes to the ornithocheirids.
- The **dsungaripterids**, whose forceps-like jaw tips and broad, rounded back teeth indicated a likely diet of shelled molluscs.
- The possibly frugivorous **tapejarids** and their possible close relatives the **thalassodromeids**, with their enormous head crests.
- The **azhdarchids**, that included the last and largest pterosaurs, with wingspans likely exceeding 10m and a long, stiff neck that may have enabled terrestrial carnivory.

Pterosaur origins

When pterosaurs first appear in the fossil record at the end of the Triassic, they are already 'fully-formed' pterosaurs – we have yet to find any obvious intermediates. This, coupled with their highly derived anatomy, has made it difficult to confidently place the group phylogenetically, beyond the clear indication of archosaur affinities (notably the antorbital fenestra). A close relationship with dinosaurs has long been suspected, though more basal placements within the **archosauriforms** have also been proposed. Recently, fairly strong evidence has emerged that a group of small-bodied, Middle-Late Triassic archosaurs called the **lagerpetids** are close pterosaur relatives, with the dinosauromorphs as the next closest. Lagerpetids had tall anterior semicircular canals, which tend to be seen in animals living in complex 3D environments, and could indicate that the group was at least partly arboreal, with potential relevance to the origin of pterosaur flight.

Pterosaur reconstructions: from bat-like to bird-like to pterosaur-like

The first reconstructions of pterosaurs treated them as very bat-like, with broad wing membranes that attached to the legs. As more soft-tissue specimens came to light, the apparent narrowness of the preserved wing membrane suggested to some that the legs may have been partially or even wholly free of the membrane. This would have compromised the facility to tension the wings, but proponents found a solution to this problem in the system of densely-packed **actinofibrils** that can be seen in the best-preserved wing membranes. It was argued that, if these could act like the feather shafts of birds – transferring aerodynamic loads directly to the wing spar – active tensioning of the wing would have been unnecessary. The legs could therefore have been free, permitting bipedal terrestrial locomotion.

While popular for a time, various lines of evidence cast serious doubt on this bird-like reconstruction. First, the discovery of large numbers of pterosaur trackways has shown conclusively that they were quadrupedal on the ground. Also, in the rare specimens that preserve the soft tissues of the proximal wing, it's clear that the trailing edge of the main wing membrane ran to the ankle across a wide taxonomic spread of pterosaurs – wide enough to indicate that this was likely a universal condition. Some of the basal pterosaur specimens of this kind – notably the Jurassic *Sordes* from Kazakhstan – also preserve a broad membrane between the legs, supported by the fifth toe. Added to this, from a mechanical point of view, the actinofibrils were too thin (the thickness of a human hair) to directly transfer loads to the wing spar – some active tensioning of the wing was essential.

Ruling out the bird-like reconstruction does not mean we have simply returned to the old bat-like reconstruction. The actinofibrils are unique to pterosaurs, and their presence only in the distal part of the wing membrane (which may explain why the distal wing tends to be preserved more often than the proximal) indicates that the wing was functionally heterogeneous. Exceptionally-preserved specimens have been found with muscle fibres within the wing membrane (a characteristic shared with bats), and parts of the wing may even have been inflatable. Possibly related to this, endocasts of pterosaur brains show that they had large **floccular lobes** of the cerebellum. Some workers suggest that these areas may have been responsible for integrating proprioceptive sensory input from the wings and eliciting an appropriate response from the muscles, including the intrinsic wing muscles. If so, this would be a unique role for the floccular lobes – they are usually concerned with gaze stabilisation.

Pterosaur flight

Flight was the most noteworthy aspect of pterosaur biology, but also one of the most difficult to comprehend. The pterosaur flight apparatus was, after all, unique (sails being the closest modern equivalent). Old ideas that pterosaurs were obligate gliders have been largely rejected: both skeletal anatomy and landing tracks indicate that at least smaller pterosaurs were fully capable of flapping

flight. What's more, they were probably able to fly at an extraordinarily young age: in marked contrast to birds and bats, the pterosaur wing finger exhibited near-isometric growth.

An underappreciated aspect of the pterosaur flight apparatus is how uncontrollable it seems to be. Lacking any rigid struts running from the wing spar to the trailing edge, there wasn't much a pterosaur could do to independently change the angle of attack of its wings. The pteroid-controlled forewing or the intrinsic wing muscles – acting in concert with the actinofibrils – may have played an important role in providing alternative control solutions, and in ameliorating a dangerous aerodynamic instability (**luffing**) that afflicts sails at low angles of attack.

An animal's mode of flight has a big impact on its ecology – an impact that can be investigated using principal component analyses of aerodynamically relevant morphological parameters such as mass and wing area. When examined in this way, many of the giant pterosaurs come out as marine soarers, with frigate birds likely having the most similar flight ecology among today's flyers. The largest pterosaurs of all such as Quetzalcoatlus, however, have a wing form that some think would have suited it more for soaring in thermals over land. Improbable as it may seem, analyses of bone strength in this taxon show that it probably could have taken off from the ground, although doubt has been cast on its thermal soaring ability.

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Highly recommended references are asterisked.

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